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An adaptive testing strategy for efficient utilization of healthcare resources during an epidemic

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ARTICLE INFO

Keywords:

Epidemic control

Adaptive testing

Efficient healthcare usage

ABSTRACT

Lockdowns are found to be effective against rapidly spreading epidemics like COVID-19. Two downsides to strategies rooted in social distancing and lockdowns are that they adversely affect the economy and prolong the duration of the epidemic. The extended duration observed in these strategies is often due to the under-utilization of medical facilities. Even though an under-utilized health care system is preferred over an overwhelmed one, an alternate strategy could be to maintain medical facilities close to their capacity, with a factor of safety. We explore the practicality of this alternate mitigation strategy and show that it can be achieved by varying the testing rate. We present an algorithm to calculate the number of tests per day to maintain medical facilities close to their capacity. We illustrate the efficacy of our strategy by showing that it reduced the epidemic duration by 40% in comparison to lockdown-based strategies.

1. Introduction

The COVID-19 pandemic has caused millions of deaths and economic hardships throughout the world. The primary fear during an epidemic is that the number of people requiring medical care would far exceed the available healthcare facilities (Fig. 1a). This scenario would cause high mortality due to the unavailability of basic medical care even when the disease is not fatal. This is especially the case with COVID-19 as most infected individuals, close to 90%, have mild-to-moderate symptoms. To prevent fatalities due to paucity of healthcare resources, many countries had imposed lockdowns. The objective of such interventions is to decrease the disease-spreading-rate so that the number of infected requiring medical care is always less than the available healthcare resources (Fig. 1a). This also helps us bide time, like in the case of COVID-19, to develop a vaccine. We have seen that the strategy to lockdown has been mostly successful around the globe. However, the only drawbacks of this strategy are the prolonged duration of the epidemic and the economic hardships resulting from it. The extended duration is primarily because the available healthcare resources are under-utilized for the most part of the epidemic. They are maximally utilized only at the peaks of each wave as shown in Fig. 1a. An obvious alternate strategy is wherein the healthcare capacity is completely utilized, with a factor of safety, during the epidemic (solid pink curve in Fig. 1a). Note that the gap between the dashed line corresponding to healthcare capacity and plateau of the

proposed-strategy-curve denotes the factor of safety. This strategy has its limitations and may not apply to all epidemics. For example, it may not be prudent to employ this strategy for a disease which cannot be treated with adequate hospital care and requires a new vaccine or drug to prevent fatalities. But, it could be relevant in case of epidemics with mild symptoms in a large percentage of the population. Here, we present a preliminary exploration of this alternate strategy to shorten the duration of the epidemic.

The proposed strategy (pink curve in Fig. 1a) requires intricate control of the epidemic, which may not be possible using gross interventions such as lockdowns. We hypothesized that this could be achieved by varying the testing rate. Most previous studies with testing and tracing have focused on simulating their effects at various uniform rates and recommending the suitable regime among them (Kerr et al., 2021; Cherian et al., 2021; Aleta et al., 2020; Sturniolo et al., 2021). It may be noted that the final recommendation from these studies is a fixed rate of testing or tracing, which typically under-utilize the medical facilities (Aleta et al., 2020). In contrast, the possibility of an epidemic control using a time-varying testing rate has not been explored adequately. The only study that is reported in the literature towards this is an adaptive testing method to keep the number of infected individuals at every instant of the epidemic a constant (Acemoglu et al., 2021). We propose another adaptive testing strategy for efficient utilization of healthcare resources (pink curve in Fig. 1a). We control

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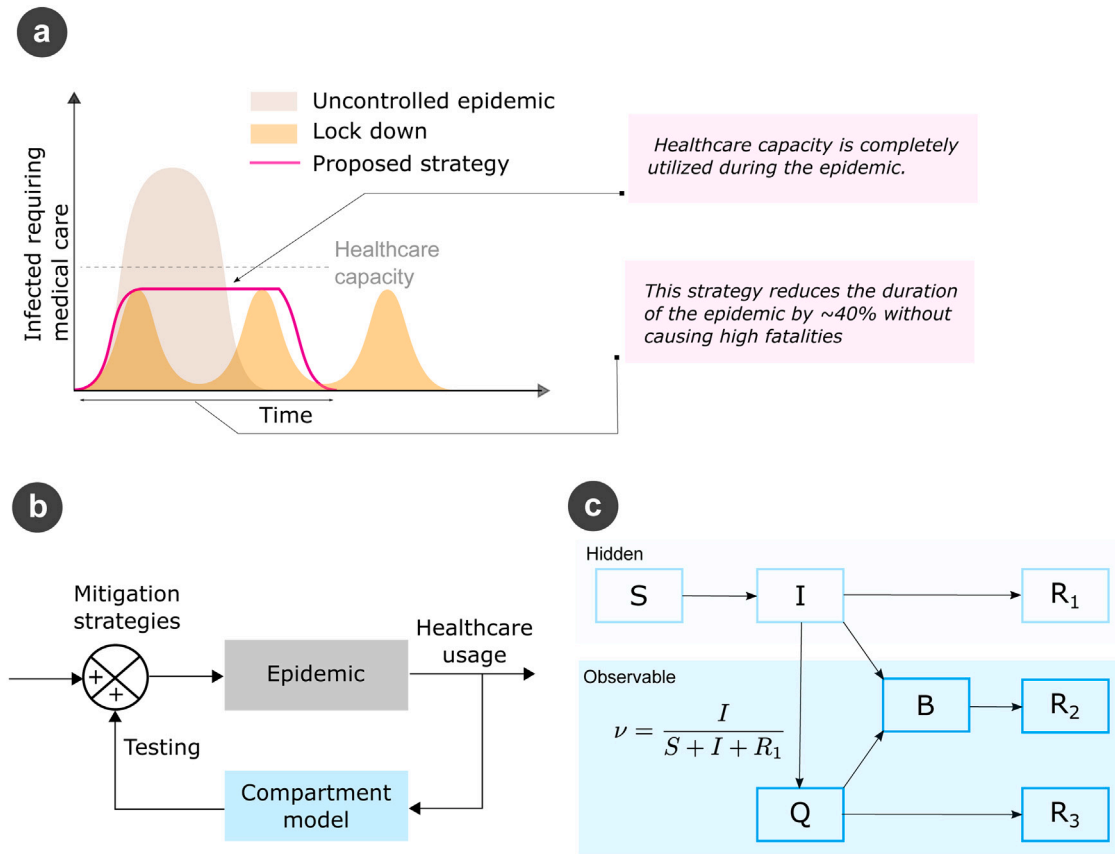


Fig. 1. Proposed strategy for efficient utilization of healthcare resources. (a) In an uncontrolled epidemic, the number of infected requiring medical care exceeds the available healthcare facilities leading to high fatalities. Alternating lockdowns and relaxations reduce fatalities by ensuring adequate medical facilities but increases the duration of the epidemic and result in multiple epidemic waves. We propose a strategy wherein the usage of healthcare facilities is maintained at its capacity for reducing the duration of the epidemic while ensuring adequate medical facilities. (b) To realize this strategy, we propose a model-predictive control technique, wherein testing rate (number of tests per day) is used as a control signal. The epidemic is modelled as a simple compartment model (c) with the following compartments; S - susceptible, I - infectious, B - infected requiring the critical healthcare resource, Q - quarantined, and $R_{1,2,3}$ - recovered from their respective connected compartments. S , I and R_1 are individuals outside the medical system, hence hidden variables. Q , B , R_2 , and R_3 are within the medical system and are hence observable variables. By testing random individuals from the hidden population, we identify infectious individuals and quarantine them ($I \rightarrow Q$), thereby controlling the spread of infection. Test positivity is also an observable variable. The testing rate required to maintain the usage of healthcare facilities at its capacity is expressed in terms of the observable variables and fed back into the epidemic to control it.

the epidemic by using testing as a control signal, which is modulated in real-time based on the current occupancy and rate of increase of occupancy of the medical facilities. In other words, we intensify testing when the healthcare facilities approach their peak capacity and relax testing when otherwise. Please note that this strategy does not restrict additional individuals from getting tested at any time. This method can be adopted by local administrative units such as district, town or county with a sizeable population and associated medical facilities. Furthermore, this strategy can be even more effective by combining proven techniques like pooled testing and contact tracing. We discuss these possible extensions of the model briefly in Section 4.

We use a model-predictive control technique for realizing this strategy (Fig. 1b). Testing rate, the number of tests per day (p), is calculated using a simplified compartment model that contains quarantine (Q) and critical healthcare resource (B) compartments (Fig. 1c). The critical medical resource is the one that gets exhausted first, which could be ICU beds, beds with oxygen facility or ventilators. The occupancy of the compartments within the healthcare system, B , Q , R_2 , and R_3 , are known and, therefore, are observable variables. The occupancy of the compartments outside the healthcare system, S , I , and R_1 , are unknown and therefore hidden variables. To control the epidemic, the infectious individuals (I) among the hidden population should be identified and quarantined. This is achieved by testing random individuals from the hidden population. In the case of random testing among the hidden population ($S + I + R_1$), the test positivity, ν , will be equal to

the fraction of infectious individuals in the hidden population. Hence, this fraction is also an observable variable. We derive an expression for p required to maintain B at B_{max} in terms of the current values of the observable variables (see Methods section). This expression and the values of the observable variables at the end of each day are used to calculate the number of tests for the next day. By using this testing rate, the epidemic progression for the next day is simulated, and the process is repeated until the end of the epidemic. Epidemic progression can be adequately simulated only using complicated models such as network or agent-based models, i.e., in the block diagram shown in Fig. 1b, the epidemic block will contain a network or agent-based model. The testing rate will be calculated using the compartment model block (Fig. 1b), which contains the model shown in Fig. 1c. However, to demonstrate the feasibility of our strategy, here we have simulated the epidemic evolution (epidemic block in Fig. 1b) using the same compartment model shown in Fig. 1c.

2. Methods

For the compartment model depicted in Fig. 1c, we assume that the rate of change of susceptible, $\dot{S}$, is directly proportional to S and I (like typical SIR models Kermack and McKendrick, 1927), i.e.,

$$\dot{S} = -\beta SI \quad (1)$$

Thus, S decreases by βSI and causes I to increase. By taking p as the number of tests per day and ν as the test positivity (that is also equal to

$I/(S+I+R_1)$, $p\nu$ individuals enter quarantine Q from I . We denote the number of days for recovery from the I and Q compartments as $1/\alpha_1$ and from B as $1/\alpha_2$. Hence, the recovery rate from I and Q are $\alpha_1 I$ and $\alpha_1 Q$, respectively. The recovered from the I , B , and Q compartments are R_1 , R_2 , and R_3 , respectively. The ratio of infected requiring critical resources to those who show mild symptoms is assumed to be δ , i.e., while $\alpha_1 I$ individuals recover to R_1 , $\delta\alpha_1 I$ need critical resources in B . Similarly, $\delta\alpha_1 Q$ move to B while $\alpha_1 Q$ recover to R_3 . Thus, we can write

$$\dot{I} = \beta SI - p\nu - \delta\alpha_1 I - \alpha_1 I \quad (2)$$

$$\dot{Q} = p\nu - \delta\alpha_1 Q - \alpha_1 Q \quad (3)$$

By defining the outflow of individuals from I and Q to B as $\delta\alpha_1 I$ and $\delta\alpha_1 Q$ respectively, we get the aforementioned definition of δ ($= R_2|_{I=\infty} / (R_1|_{I=\infty} + R_3|_{I=\infty})$). Note that quarantined individuals (Q) are assumed not to transmit the infection. It is now straightforward to see that the remaining governing equations of the model are

$$\dot{B} = \delta\alpha_1 (I + Q) - \alpha_2 B \quad (4)$$

$$\dot{R}_1 = \alpha_1 I \quad (5)$$

$$\dot{R}_2 = \alpha_2 B \quad (6)$$

$$\dot{R}_3 = \alpha_1 Q \quad (7)$$

It may be noted that only Q , B , R_2 and R_3 compartments and the test positivity, ν , are observable but S , I and R_1 are hidden (Fig. 1c).

Our strategy is to force B to trace a desired function using testing. The desired function was chosen as a sigmoid that starts at zero and saturates at B_{max} .

$$B_s = \frac{B_{max}}{1 + e^{-c(t-t_h)}} \quad (8)$$

where c controls the rise of the sigmoid. For B to trace B_s , $\dot{B}$ at a given value of B should be equal to $\dot{B}_s$, which is given by

$$\dot{B}_s = c B_{max} \frac{e^{-c(t-t_h)}}{[1 + e^{-c(t-t_h)}]^2} \quad (9)$$

$$= c \frac{B_s^2}{B_{max}} \left(\frac{B_{max}}{B_s} - 1 \right) \quad (10)$$

To converge $\dot{B}$ to $\dot{B}_s$, we define a desired $\ddot{B}$, $\ddot{B}_d$, as the difference between them

$$\ddot{B}_d = \ddot{B} - \ddot{B}_s \quad (11)$$

where the time for converging $\dot{B}$ to $\dot{B}_s$ is assumed to be one day. Expression for $\ddot{B}$ can be obtained by differentiating the expression for $\dot{B}$ in the compartment model, Eq. (4)

$$\ddot{B} = \delta\alpha_1 (\dot{I} + \dot{Q}) - \alpha_2 \dot{B} \quad (12)$$

Substituting for $\dot{I}$ and $\dot{Q}$ by using Eqs. (2) and (3),

$$\ddot{B} = \delta\alpha_1 \beta SI - \delta\alpha_1 (\delta\alpha_1 + \alpha_1 + \alpha_2)(I + Q) + \alpha_2^2 B \quad (13)$$

Substituting the desired $\ddot{B}_d$ from Eq. (11) for $\ddot{B}$ and re-arranging we get the desired infection rate,

$$\dot{S}_d = (\beta SI)_d = \frac{\ddot{B}_d - \alpha_2^2 B}{\delta\alpha_1} + (\delta\alpha_1 + \alpha_1 + \alpha_2)(I + Q) \quad (14)$$

By achieving this desired infection rate, $\dot{B}$ converges to $\dot{B}_s$ thereby tracing the sigmoid. The desired rate of change of infection rate,

$$\ddot{S}_d = (\beta \dot{S} I)_d = \frac{(\beta SI)_d - (\beta SI)}{\Delta t} \quad (15)$$

where βSI is the current rate of infection and Δt is a time period chosen for smooth convergence of current infection rate $\dot{S}$ to desired infection rate $\dot{S}_d$. Rate of change of infection rate can be written as

$$(\beta \dot{S} I) = \beta (\dot{S} I + S \dot{I}) \quad (16)$$

$$= \beta (\dot{S} I + S (\beta SI - p\nu - (\delta\alpha_1 + \alpha_1) I)) \quad (17)$$

Substituting for $(\beta \dot{S} I)$ from Eq. (15) and re-arranging

$$p = \frac{1}{\nu} \left[\left(1 - \frac{I}{S} \right) \beta SI - (\delta\alpha_1 + \alpha_1) I - \frac{(\beta SI)_d - (\beta SI)}{\beta S \Delta t} \right] \quad (18)$$

By testing at the rate given by the above equation, the infection rate ($\dot{S}$) converges to the desired infection rate ($\dot{S}_d$), which in turn converges the rate of increase of critical resource usage ($\dot{B}$) to the gradient of the sigmoid ($\dot{B}_s$). This equation cannot be applied directly because it contains hidden variables S and I . However, S and I can be expressed in terms of observable variables. For expressing I in terms of observable variables, we use test positivity and assume that the total population does not change during the epidemic.

$$\nu = \frac{I}{S + I + R_1} \quad (19)$$

$$= \frac{I}{N_1} \quad (20)$$

where $N_1 = S + I + R_1$ is the number of people in the hidden compartments. Since the total population is assumed to remain constant during the epidemic,

$$N = (S + I + R_1) + (Q + B + R_2 + R_3) \quad (21)$$

$$\Rightarrow N_1 = N - (Q + B + R_2 + R_3) \quad (22)$$

Now, N_1 is expressed in terms of the observable variables and therefore $I = \nu N_1$ is also obtained in terms of the observable variables. To express S in terms of observable variables, we take the derivative of ν by differentiating Eq. (20).

$$\dot{\nu} = \frac{N_1 \dot{I} - I \dot{N}_1}{N_1^2} \quad (23)$$

Substituting for $\dot{I}$ from the compartment model, Eq. (2), and re-arranging we get an expression for the infection rate,

$$\beta SI = p\nu + (\delta\alpha_1 + \alpha_1) I + \frac{N_1^2 \dot{\nu} - I \dot{N}_1}{N_1} \quad (24)$$

By substituting $I = \nu N_1$ in these equations and rearranging, S can be expressed in terms of observable parameters.

We have implemented these relations as follows. Testing commences only after a specific percentage of B is occupied. For our simulations we have used 20% of B as the threshold for starting testing. At the end of each day, n , we compute the number of tests required to be performed on the next day, $n+1$, using the above relations. We first estimate the desired rate of change of $\dot{B}$ at n th day ($\dot{B}_{d,n}$) using Eq. (11)

$$\dot{B}_{d,n} = c \frac{B_n^2}{B_{max}} \left(\frac{B_{max}}{B_n} - 1 \right) - \dot{B}_n \quad (25)$$

where B_n is the usage of critical resource on the n th day and $\dot{B}_n$ is the rate of change of critical resource usage. $\dot{B}_n$ was obtained by calculating the slope of a straight line fit to B for the previous three days. Next, we calculate the desired infection rate using Eq. (14)

$$(\beta SI)_{d,n} = \frac{\dot{B}_{d,n} - \alpha_2^2 B_n}{\delta\alpha_1} + (\delta\alpha_1 + \alpha_1 + \alpha_2)(I_n + Q_n) \quad (26)$$

This desired infection rate is used to calculate the number of tests for the $n+1$ th day using Eq. (18)

$$p_{n+1} = \frac{1}{\nu_n} \left[\left(1 - \frac{I_n}{S_n} \right) \beta S_n I_n - (\delta\alpha_1 + \alpha_1) I_n - \frac{(\beta SI)_{d,n} - (\beta S_n I_n)}{\beta S_n \Delta t} \right] \quad (27)$$

where $\Delta t = 10$ days, $I_n = \nu_n N_{1,n}$ and S_n is obtained from Eq. (24)

$$S_n = \frac{p_n \nu_n}{\beta I_n} + \frac{\delta\alpha_1 + \alpha_1}{\beta} + \frac{N_{1,n}^2 \dot{\nu}_n - I_n \dot{N}_{1,n}}{\beta I_n N_{1,n}} \quad (28)$$

It may be noted that this Δt is not the step size of our ODE integration. The step size for integration is much smaller than 1 day and was

Table 1

Parameters used for the adaptive-testing strategy.

Parameter	Value
N	1.5 million
B_{max}	0.2 per 1000
R_0	2.0
$1/\alpha_1$	7 days
$1/\alpha_2$	15 days
δ	0.02
c	0.08
p_{max}	0.45 million per day

automatically chosen by the ODE solver (ode45) in MATLAB. By using Eqs. (27) and (28) at the end of each day, the number of tests for the next day was updated. This updated testing rate was used to simulate the epidemic for the next day and the process was repeated until the end of the epidemic. Minimum and maximum tests per day were set at 0 and 0.45 million (0.3N) respectively to avoid wild fluctuations in testing. c was chosen to be 0.08. The algorithm was implemented in MATLAB (Supplementary Materials).

This strategy was evaluated for epidemic parameters corresponding to the first wave of COVID-19 epidemic in India (Table 1). The total population of the state of Goa was considered, $N = 1.5$ million. The critical resource was assumed to be ICU beds and its availability was obtained from an official government website, $B_{max} = 0.2$ per 1000 (Goa, 2021). Reproduction number, $R_0 = 2.0$ (Ferretti et al., 2020), was used to calculate $\beta = R_0\alpha_1(1+\delta)/N \approx R_0\alpha_1/N$ since $\delta \ll 1$. $1/\alpha_1 = 7$ days and $1/\alpha_2 = 15$ days were chosen from the recovery periods for asymptomatic and severely infected individuals reported in Childs et al. (2021) and Hazra et al. (2022). From Kerr et al. (2021) and Hazra et al. (2022), we obtained $\delta \approx 0.017$ by averaging over all age groups and assuming that both asymptomatic and mild infections do not require the critical resource. This was rounded off to obtain $\delta = 0.02$.

In the previous strategy, the contact rate, β , was assumed to be known even though its a hidden variable. Since β is difficult to estimate during the epidemic, we further derived a method which does not require the value of β to implement our adaptive-testing strategy. The rate of change of infection rate from Eq. (17) can be written as

$$(\beta\dot{S}) = (\beta\dot{S})I + (\beta S)\dot{I} \quad (29)$$

$$= (\beta\dot{S})I + \beta S\{\beta SI - p\nu - (\delta\alpha_1 + \alpha_1)I\} \quad (30)$$

where we keep βS as one term and do not split it into β times S like we had done previously. Rearranging we get

$$p = \frac{1}{\nu} \left[\beta SI - (\delta\alpha_1 + \alpha_1)I - \frac{(\beta\dot{S})I - (\beta\dot{S})I}{\beta S} \right] \quad (31)$$

βS was obtained from the infection rate Eq. (24) at the end of each day as

$$(\beta S)_n = \frac{p_n \nu_n}{I_n} + \delta\alpha_1 + \alpha_1 + \frac{N_{1,n}^2 \dot{\nu}_n - I_n N_{1,n}}{I_n N_{1,n}} \quad (32)$$

$(\beta\dot{S})$ was obtained by calculating the slope of a straight line fit to βS for the previous five days. These values for $(\beta S)_n$ and $(\beta\dot{S})_n$ can be used to update p , Eq. (31),

$$p_{n+1} = \frac{1}{\nu_n} \left[(\beta\dot{S})_n - (\delta\alpha_1 + \alpha_1)I_n - \frac{(\beta\dot{S})_{d,n} - (\beta\dot{S})_n}{(\beta S)_n \Delta t} + \frac{(\beta\dot{S})_n I_n}{(\beta S)_n} \right] \quad (33)$$

3. Results

We first evaluate the performance of our adaptive testing algorithm in controlling the epidemic (Fig. 2a–c). We assume the following parameters for the epidemic (Table 1): total population $N = 1.5$ million (approximate population of the state of Goa, India); $B_{max} = 0.2$ per 1000 (Goa, 2021); $R_0 = 2.0$ (Ferretti et al., 2020); ratio of

infected requiring critical resource to those with mild symptoms, $\delta = 0.02$ (Hazra et al., 2022); average duration for recovery from I and Q , $1/\alpha_1 = 7$ days (Kerr et al., 2021; Hazra et al., 2022); and average duration for recovery from B , $1/\alpha_2 = 15$ days (Hazra et al., 2022). The adaptive testing strategy could maintain B at B_{max} throughout the epidemic. The strategy is imposed when B increases beyond 20% of B_{max} , corresponding to the point where the colour of the curve changes from brown to pink in Fig. 2a. B increases smoothly to B_{max} and further stays constant at that level (pink curve in Fig. 2a). The epidemic tails off after herd immunity has been achieved (grey curve in Fig. 2a). The corresponding variations in the number of infectious (solid curves) and quarantined (dashed curves) individuals are shown in Fig. 2b. The number of infectious individuals reaches a peak of less than 5 per 1000 just before the epidemic declines. The variation in tests per day required to achieve this is shown in Fig. 2c. When testing commences, there is an instability that rapidly stabilizes within 20–30 days. This can be seen from the fluctuations in testing rate shown in the inset in Fig. 2c. These fluctuations are because the testing rate is inversely proportional to the test positivity, ν , (refer to Eq. (18) in the Methods Section), which changes rapidly due to quarantining of infectious individuals (dashed curve in Fig. 2c). After stabilization, the testing rate reaches a maximum and further decreases smoothly to zero. The maximum number of quarantined individuals is around 2.4 per 1000, early in the epidemic, around the time when the testing reaches a maximum after stabilization (dashed curve in Fig. 2b).

Unmitigated epidemic, without any interventions, is shown in Fig. 2d. Here, B increased to more than 25 times (≈ 5 per 1000 people) its capacity (B_{max} assumed to be 0.2 per 1000 people), suggesting high fatalities (Fig. 2d). Next, we compare our approach with strategies such as lockdown and constant testing rate (Fig. 2e and f). These mitigation strategies are imposed when B goes above 20% of B_{max} and are discontinued when B goes below 5% of B_{max} (yellow curves in Fig. 2e and f represent the phase when the strategies are active). When the mitigation strategies are relaxed, B increases (brown curves in Fig. 2e and f) and the mitigation strategies were reimposed when B crossed 20% of B_{max} . This gave rise to multiple waves of the epidemic (Fig. 2e and f). The parameters of the epidemic are same as that assumed for the adaptive testing strategy. In case of lockdowns, R_0 was decreased to ensure that the B does not increase beyond B_{max} in that wave. By restricting R_0 to 1.09, 1.31, and 1.62 for the successive waves, B at the peak of each wave was approximately equal to B_{max} (Fig. 2e). The increasing values of R_0 for successive waves implies easing of lockdown and social-distancing measures as the epidemic progresses. This is tenable because the infection rate for a given number of infectious individuals is lower in successive waves since the number of susceptible individuals decreases as the epidemic progresses. After the third wave, the epidemic winds down (grey curve in Fig. 2e). The total duration of lockdowns (yellow curves in Fig. 2e) are around 1300 days with brief periods of relaxation in between (brown curves in Fig. 2e). Similar results were obtained with a constant testing rate strategy (Fig. 2f). By testing at a constant rate of 139, 110, 78, 45, and 12 per 1000 people per day in five waves, B could be restricted below B_{max} at the peak of each wave. The total duration of this mitigation strategy is around 1500 days.

In comparison to lockdown and constant testing rate strategies, wherein the critical resource is completely utilized only at the peak of each wave, the adaptive testing rate strategy maintains B at B_{max} throughout the epidemic and therefore enables efficient utilization of healthcare resources. Better utilization of medical resources reduces the duration of the epidemic. In the adaptive testing strategy, the duration of the epidemic is ≈ 800 days, which is around 40% lower than other strategies. It may be noted that the total number of individuals requiring critical care is approximately the same between these strategies (difference $< 10\%$). This is because herd immunity is achieved after S decreases below a threshold, which depends only on R_0 and the number of individuals requiring critical care is assumed to be a

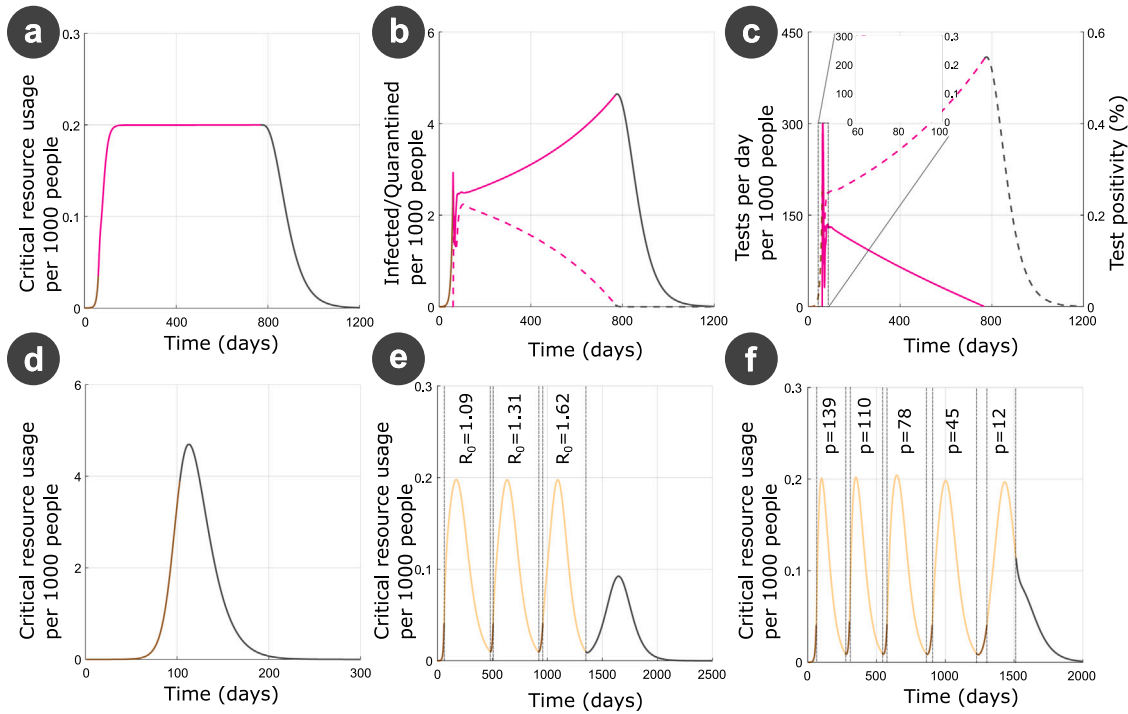


Fig. 2. Comparison between epidemic mitigation strategies. (a–c) Adaptive testing. Evolution of critical resource usage, $(B/N) \times 1000$ is shown in (a), infected, $(I/N) \times 1000$ (solid curves), quarantined, $(Q/N) \times 1000$ (dashed curves) are shown in (b), tests per day per 1000 people, $(p/N) \times 1000$ (solid curves), and test positivity, v (dashed curves) are shown in (c). (d) Uncontrolled epidemic. (e) Lockdown controlled epidemic. R_0 is controlled through lock downs or social-distancing norms and (f) Epidemic controlled by a constant testing rate. Colours indicate different regimes during the epidemic - uncontrolled (brown), mitigation strategies applied (pink in a–c and yellow in e and f), and epidemic winding down (grey).

fixed percentage of the total infected individuals during the epidemic. Hence, the adaptive testing strategy decreases the duration of the epidemic without higher hospitalization rates by efficient utilization of the available healthcare resources. Essentially, we have shown that the usage of medical facilities can be set to a constant value throughout the epidemic by using our strategy. The local health authorities can choose that constant value to be 70, 80, 90 or 100% of the actual capacity. This can give a safety factor for any of the unforeseen circumstances such as breakdown or maintenance of the medical facilities and critical incidents from other diseases.

Furthermore, we tested the performance of our strategy by varying the epidemic parameters such as B_{max} (Fig. 3a–c), R_0 (Fig. 3d–f), δ (Fig. 3g–i), and $1/\alpha_2$ (Fig. 3j–l). In all cases, our algorithm could maintain B at B_{max} . Different B_{max} (0.15 - blue, 0.2 - green, 0.25 - red, and 0.3 per 1000 people - black curves in Fig. 3a–c) correspond to places with different availability of medical facilities. Higher B_{max} can sustain higher I (solid curves in Fig. 3b) thereby achieving herd immunity earlier and consequently reducing the duration of the epidemic (Fig. 3a). Higher B_{max} does not necessitate higher testing rates. The peak testing rate after stabilization is equal for different B_{max} (solid curves in Fig. 3c) and the duration of epidemic is smaller for larger B_{max} . Different R_0 (1.75 - blue, 2.0 - green, 2.25 - red, and 2.5 - black curves in Fig. 3d–f) correspond to viral strains with different transmissibility. For example, the delta strain of COVID-19 had higher R_0 than previous strains (Liu and Rocklöv, 2021). Higher R_0 necessitated higher testing rates (Fig. 3f) and the epidemic lasted longer (Fig. 3d and e). The increase in the duration of the epidemic is because the threshold of S below which the epidemic declines is lower for higher R_0 . Different δ (0.005 - blue, 0.01 - green, 0.015 - red, and 0.02 - black curves in Fig. 3g–i) correspond to strategies such as vaccinations and isolation of the vulnerable. Alterations in δ do not alter testing rates (Fig. 3i) and reduction in δ can drastically decrease

the duration of the epidemic (Fig. 3g). Different $1/\alpha_2$ (11 - blue, 13 - green, 15 - red, 17 days - black curves in Fig. 3j–l) correspond to treatment regimes that can alter the time spent in the critical resource till recovery. For instance, steroids are known to decrease stay in ICUs for COVID-19. Changes in $1/\alpha_2$ does not change the peak testing rate (Fig. 3l) whereas reducing $1/\alpha_2$ can decrease the duration of the epidemic (Fig. 3j). As mentioned before, maximum testing rate, slope of the sigmoid (c) and the point of initiation of the testing strategy were chosen as 300 per 1000, 0.08 and 20% of B_{max} . A discussion on the choice of these parameters is given in the supplementary information. We also chose window sizes = 3 days for calculating the rate of change of B , v and N_1 and time intervals for converging the rate of change of $B = 1$ day (Eq. (11)) and infection rate = 10 days, (Eq. (15)). We performed numerical experiments to understand the effect of varying these window sizes and time intervals. When the testing commences, B and v are increasing rapidly whereas N_1 is decreasing rapidly. A larger window size, for example 7 days, will underestimate the rate of change of these variables and cause large fluctuations in p . Interestingly, even with such large fluctuations, the evolution of B is mostly unaffected; B increases to B_{max} and then remains constant until the end of the epidemic. A shorter time interval, for example 1–5 days in Eq. (15), will require a larger number of tests and causes large fluctuations in testing whereas a larger time interval, for example 5 days in Eq. (11) or 100 days in Eq. (15), will result in B overshooting B_{max} .

The SIR model assumes that the individuals belonging to S , I and R compartments are uniformly distributed among the population, which is a simplification. Furthermore, the epidemic will not spread with a uniform R_0 due to variations in the climate, large gatherings, school opening and closing and super spreaders. To evaluate the performance of our method under such uncertainties and noise in the real world, we varied the contact rate during the epidemic. R_0 was varied according to a normal distribution. We fixed a mean value and standard deviation,

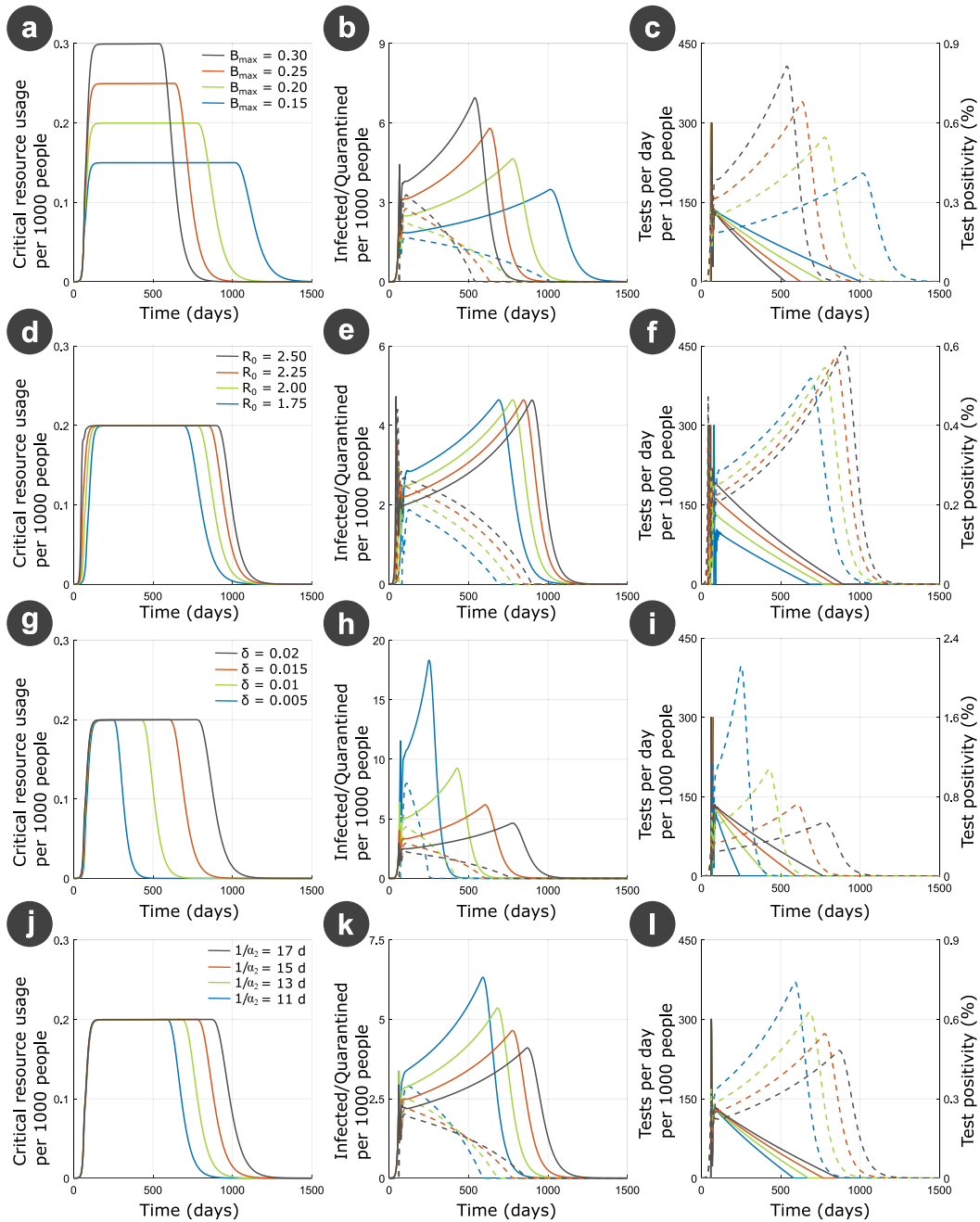


Fig. 3. Performance of the adaptive testing strategy when the epidemic parameters are varied. (a–c) $B_{\max}$, (d–f) R_0 , (g–i) δ and (j–l) $1/a_2$. When any of these parameters were varied, the other parameters were held constant at the following values ($B_{\max}/N \times 1000 = 0.2$, $R_0 = 2.0$, $\delta = 0.02$, $1/a_2 = 15$ days). Left column shows critical resource usage, $(B/N) \times 1000$; middle column shows infected $(I/N) \times 1000$ (solid lines) and quarantined $(Q/N) \times 1000$ (dashed lines); and the right column shows testing rate, $(p/N) \times 1000$ (solid lines) and test positivity v (dashed lines).

drew samples from that distribution and obtained R_0 for each day of the epidemic. From the R_0 , we calculated the spreading rate β for each day, which was used to simulate the epidemic for that day. In such a scenario, unlike the previous cases, we cannot assume that we know the hidden variable β . Hence, we used Eq. (33), which does not require the value of β to implement the adaptive-testing strategy. We evaluated the performance of this strategy by varying the standard deviation of the normal distribution ($\mu = 2.0$ and $\sigma = 0.1, 0.3, 0.5$, Fig. 4). The strategy was able to stabilize the critical resource usage at its maximum throughout the epidemic. Larger fluctuations in R_0 necessitated larger fluctuations in the testing rate (Fig. 4c) in order to stabilize B at $B_{\max}$ (Fig. 4a). Even though there are fluctuations in the

epidemic parameters, their general trend, in comparison to epidemic with constant R_0 (Figs. 2, 3), remained the same.

4. Discussion

Our intention is to enrich the available techniques for epidemic management, which can be evaluated by the local healthcare authorities after a thorough comparison among them. As shown from our simulations, the major benefits of our strategy are the effective utilization of medical facilities and subsequent reduction in the epidemic duration with minimal economic impact to individuals. On the other hand, there is a risk of higher deaths in our strategy in comparison to imposing lockdowns until a vaccine is developed. In the latter

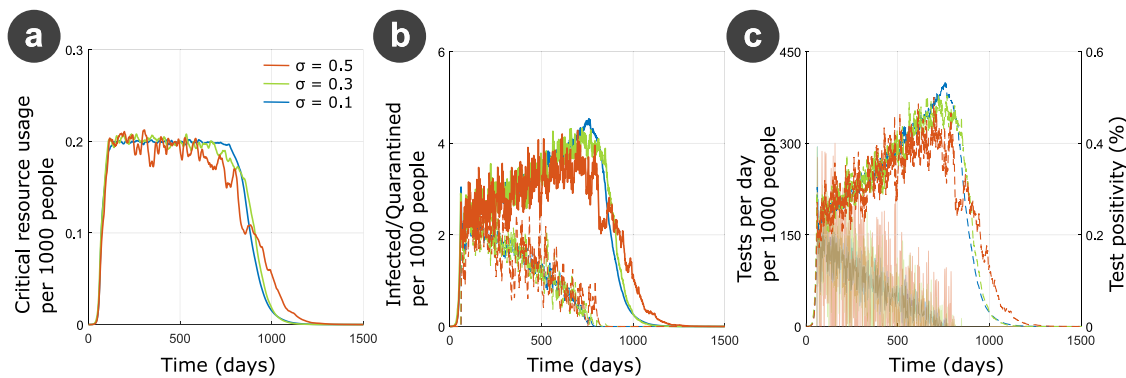


Fig. 4. Performance of the adaptive testing strategy for epidemics with varying R_0 . R_0 was varied during the epidemic according to a normal distribution with mean = 2.0 and standard deviation = 0.1 (blue), 0.3 (green) and 0.5 (red). Left column shows critical resource usage, $(B/N) \times 1000$; middle column shows infected, $(I/N) \times 1000$ (solid lines) and quarantined $(Q/N) \times 1000$ (dashed lines); and the right column shows testing rate, $(p/N) \times 1000$ (solid lines) and test positivity ν (dashed lines).

strategy, it is prudent to prevent disease spreading early in the epidemic since infections post-vaccination have lower chances of fatality. By comparing these competing scenarios, the local authorities can take an informed decision in cognisance with their local conditions and constraints.

In the beginning of the COVID-19 epidemic, a few countries such as UK and Sweden had resisted nationwide lockdowns and tried to achieve herd immunity by allowing the infection to spread through the younger population who are considered to be relatively safe (Orlowski and Goldsmith, 2020). However, the rising number of cases and hospitalizations forced these countries to impose strict social-distancing measures (Claeson and Hanson, 2021; Colbourn, 2020). This was primarily because the healthcare system got stressed and there were no known methods other than social-distancing to blunt the rise in hospitalizations. The method proposed here ‘flattens the hospitalization curve’, which could have allowed these countries to continue without stringent lockdowns and thereby economic disruptions. In contrast to UK and Sweden, most of the countries imposed strict social-distancing measures until the development of vaccines and drugs to decrease the risk of severe disease and fatalities in high-risk individuals. However, as shown in Fig. 3g and h, even at lower δ and $1/\alpha_2$ if the epidemic is allowed to spread freely, the medical facilities can get overwhelmed. This too can be avoided by the adaptive testing strategy, which provides an attractive exit strategy without resorting to additional lockdowns.

We showed in Fig. 2 that by flattening the hospitalization curve, we significantly decrease the duration of the epidemic. For a realistic set of epidemic parameters, chosen based on the available COVID-19 data, we showed that the reduction in duration is 40%. We illustrated the robustness of the strategy by varying the parameters B_{max} , R_0 , δ , and $1/\alpha_2$ (Fig. 3). In other words, for a wide range of values of the epidemic parameters our strategy enables efficient utilization of medical facilities and thereby decreases the duration of the epidemic.

In practice, three of the epidemic parameters— B_{max} , δ and $1/\alpha_2$ —are controllable while R_0 is not entirely controllable since it depends on parameters such as the transmissibility of the viral strain. For example, B_{max} can be improved by building more medical facilities and vaccinations will decrease δ and R_0 . By developing new therapeutic strategies for reducing time spent in hospitals, $1/\alpha_2$ can be reduced. The results show that when we combine these measures with the adaptive testing strategy, the epidemic duration can be further decreased. Hence, the adaptive testing is a new paradigm in epidemic control, which incidentally works even effectively in conjunction with the existing pharmaceutical and non-pharmaceutical strategies.

One of the limitations of our method is the very high testing rate required, around 300 per 1000 individuals per day at its peak. In comparison, the peak testing rate in populous countries such as India and US have peaked at around 2–6 per 1000 individuals per day (Ritchie et al., 2020). Interestingly, a few smaller countries such

as Slovakia, Cyprus, and Austria have reported testing rates of 100–150 per 1000 individuals per day suggesting that the high testing rates required by our method are possible (Ritchie et al., 2020). We could further decrease the testing rates required by our method by combining it with other strategies such as pooled testing, contact tracing and testing random households instead of individuals:

- **Pooled testing** — Pooling samples can significantly decrease the number of tests required to identify infected individuals in cases of low prevalence (Dorfman, 1943). Many countries such as USA (Abdallhamid et al., 2020), Israel (Yelin et al., 2020) and Germany (Lohse et al., 2020) have employed pooled testing during the COVID-19 pandemic. In our case, test positivity $< 0.5\%$ (Figs. 2c and 3c, f, i, l), the number of tests can be reduced by a factor of 0.068 (Mutesa et al., 2021) thereby decreasing the peak testing rate to 20 per 1000 individuals per day. Such testing rates have been reported during the COVID-19 pandemic by several countries such as UK, UAE, Czech Republic, and Denmark (Ritchie et al., 2020).
- **Contact tracing** — The fundamental idea of our method is to quarantine enough infected individuals to decrease the infection spreading rate and thereby control the epidemic. In the method developed in this paper, all infected individuals are identified by testing random individuals from the population. In such a random sampling strategy, the number of infected individuals identified will be proportional to the fraction of infected in the population, which is very low. The test positivity is of the order of 0.5% (Figs. 2c and 3c, f, i, l), which means that 200 tests are needed to identify one infected individual. In contrast, among a cohort of individuals known to have been in contact with an infected person, the test positivity would be much higher. For example, a study during COVID epidemic concluded transmission probabilities of 4.7% and 10.7% among low-risk and high-risk contacts respectively in two Indian states (Laxminarayan et al., 2020), which means 10–24 tests for identifying an infected person among the contacts of another infected individual. Hence, by combining with contact tracing the testing rate can be reduced significantly.
- **Testing random households instead of random individuals** — Many studies have shown that COVID-19 spreads more rapidly indoors rather than outdoors. Moreover, people do not follow social-distancing norms such as wearing masks in their homes. Hence, when one person in a house gets infected, the chances of the entire household getting infected are very high. Therefore, one individual from a household could be representative of the entire household and if that individual tests positive, the entire household could be quarantined. This is equivalent to redefining the basic infectious unit as a household rather than an individual,

which could decrease the number of tests by the average size of a household. Latest UN data shows that average household size in different countries vary from 2.2 (France) to 8.8 (Senegal) with India and US at 4.5 and 2.5 individuals per household respectively (United Nations et al., 2019). Hence, by testing a representative individual from each randomly chosen households, we could decrease the peak testing rate to 11%–45%. However, this requires reformulating the compartment model in terms of households, which will also change the contact and recovery parameters.

Another limitation with this strategy is the unavailability of an accurate testing method that can be scaled to the levels required by this method. For example, during COVID the RATs were fast and less accurate in comparison to the RTPCR tests. Nonetheless, such quick testing methods can be used in our strategy by incorporating the false-negative rate as a scaling parameter. The number of tests will be inversely scaled by the true-positive rate (1 - false-negative rate). We have also made several simplifying assumptions since our objective was to demonstrate the viability of our strategy. We have not considered waning immunity, i.e., the recovered becoming susceptible again after a refractory period. We believe that this is tenable because the timescale for losing immunity, several months to a year, is comparable to the epidemic duration and the chances of a previously infected individual requiring critical care in a subsequent infection is very low (Medić et al., 2022). A faster immunity waning, can be considered by modifying the compartment model to a SIRS model. Another possibility is the emergence of new variants that evade the immunity caused by previous variants (Thompson et al., 2023). Such new variants would not affect our method if the immunity against previous strains can prevent severe symptoms. In case the new variant can cause severe symptoms, we might need to consider it as another epidemic along with the previous variant.

Social-distancing measures and testing are two distinct non-pharmaceutical techniques used for epidemic control. The former provides gross control over the epidemic. In this work, by using the adaptive testing strategy we showed that the latter gives fine control. We demonstrated this by recasting epidemic mitigation as a control problem wherein the testing rate is modulated in real-time. This is distinct from epidemiology wherein the focus is on predicting the long-term evolution of the epidemic. Epidemiology is extremely challenging because future predictions lose accuracy over time due to accumulation of errors and emergence of unforeseen circumstances (Scarpino and Petri, 2019; Castro et al., 2020). In contrast, control strategies typically require only short-term prediction. The control signal needs to be updated in a short time window using the most recent data. In our case, we calculate the number of tests for the next day based on the current value of the observable variables. We believe that a simple compartment model is sufficient for such short-term prediction. Here, we have used the same compartment model for simulating the evolution of the epidemic, which is a limitation of the current study. In the future, we will simulate the epidemic evolution using complicated models such as network and agent-based models. In such a case, the epidemic variables of the compartment model that are hidden should be updated in real-time by fitting the model to the latest epidemic data. As mentioned before, α_1 , α_2 , and δ are observable whereas β is hidden. β could be estimated from the variation in v . We believe that even with complex epidemic evolution models our algorithm can successfully control the epidemic because of adequate short-term predictability of compartment models.

The emerging paradigm for epidemic management is one wherein the government ensures only adequate medical facilities for the infected while delegating the responsibility of protecting oneself from infection to individuals. Our strategy enables such a paradigm without enforcing strict social-distancing measures such as lockdowns. Individuals are required to undergo testing whenever directed by the health authorities and quarantine themselves if tested positive. Hence, our strategy enables efficient epidemic mitigation with minimal intrusion into individual liberty.

CRediT authorship contribution statement

Sreenath Balakrishnan: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Safvan Palathingal:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

Acknowledgements

The authors thank Dr. Midhun S. Menon for useful discussions, Dr. Namrata M. Nilavar for helpful comments, and S.B. thanks Indian Institute of Technology Goa, India for funding (grant number - 2020/IP/SB/008).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jtbi.2023.111555>.

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